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MILITARY SPECIFICATION

TRIPHENYL BISMUTH

Inactive for new design after 22May14, and is no longer used except for replacement purposes.

This specification is approved for use by the U.S. Army Combat Capabilities Development Command – Armaments Center (CCDC – AC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements and quality assurance provisions for the manufacture and acceptance of one type of triphenyl bismuth (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to: Commander, U.S. Army CCDC – AC, ATTN: FCDD-ACE-QSA, Picatinny Arsenal, New Jersey 07806-5000 or e-mailed to usarmy.pica.ccdc-ac.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this information using ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6810

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government standards and publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM E 168 - Standard Practices for General Techniques of Infrared Quantitative Analysis
ASTM E 300 - Standard Practice for Sampling Industrial Chemicals
ASTM D 1193 - Standard Specification for Reagent Water

(Copies of these documents are available online at http://www.astm.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Physical and chemical requirements. The triphenyl bismuth shall conform to the following requirements when tested in accordance with the applicable test method.

TABLE I. Chemical and physical requirements for triphenyl bismuth.

Property	Minimum	Maximum	Value
Purity	97.0	--	4.5.1
Melting point, ˚C	76.0	78.5	4.5.2
Acetone insoluble, %	--	0.1	4.5.3
Magnesium content, %	--	0.003	4.5.4
Total halide content, %	--	0.05	4.5.5

3.2 Infrared spectrum. The infrared (IR) spectrum of the sample shall contain the major peaks shown in Figure 1when tested in accordance with 4.5.6.

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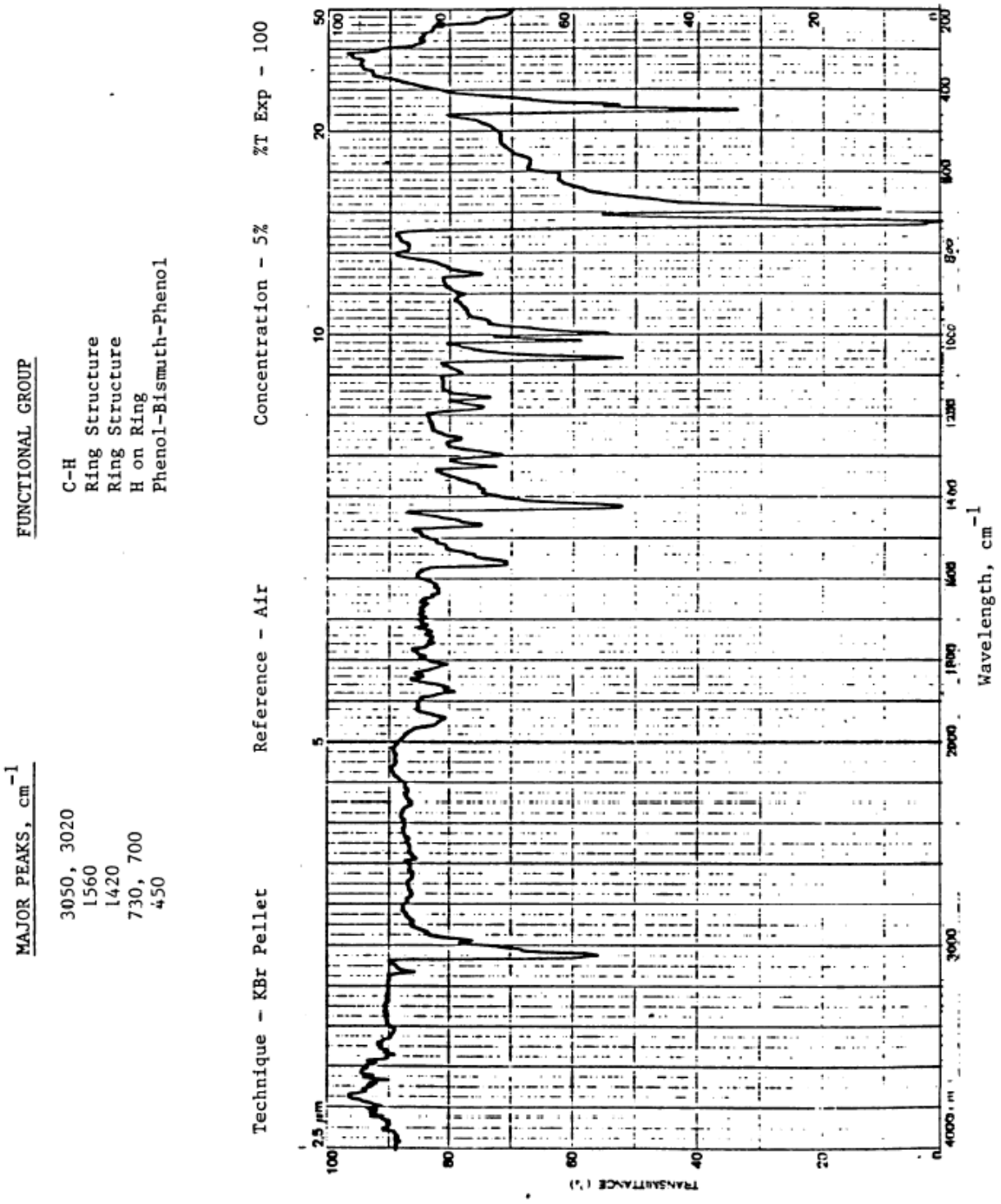


Figure 1. IR spectrum for triphenyl bismuth.

3.3 First article. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

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3.4 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. Determination of foreign materials shall be in accordance with 4.5.7.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspections specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article shall consist of ten pounds of triphenyl bismuth. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspections to performed. The sample will be subjected by the Government to any or all of the examinations and tests specified in Table II.

4.3.3 Rejection. If any sample fails to comply with any of the applicable requirements, the first article shall be rejected.

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TABLE II. First article inspection.

Examination or test	Requirement paragraph	Inspection method paragraph
Purity	3.1	4.5.1
Melting point, ˚C	3.1	4.5.2
Acetone insoluble, %	3.1	4.5.3
Magnesium content, %	3.1	4.5.4
Total halide content, %	3.1	4.5.5
Infrared spectrum	3.1	4.5.6
Workmanship	3.3	4.5.7

4.4 Quality conformance inspection.

4.4.1 Lot formation. A lot shall consist of one or more batches of triphenyl bismuth produced by one manufacturer, in accordance with the same specification, or same specification revision, under one continuous set of operating conditions. Each lot shall consist of that quantity of triphenyl bismuth that has been subjected to the same unit chemical or physical process intended to make the final product homogeneous. Lot formation shall be in accordance with MIL-STD-1916, Formation and identification of lots or batches.

4.4.2 Examination.

a. Workmanship. Examination for workmanship shall be conducted on each sample selected for testing in accordance with 4.4.3.1. If any sample fails to meet any test requirement the batch represented by the sample shall be rejected.

4.4.3 Testing. PRECAUTION: This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing the triphenyl bismuth.

4.4.3.1 Sampling. Approximately 500 grams of the composition shall be selected from each batch to be sampled using ASTM E 300 for solids. Samples shall be selected for inspection in accordance with Table III. If any sample fails to meet any test requirement the batch represented shall be rejected. Each required test shall be performed in duplicate. No composite samples shall be used. The classification of defects shall be as given in Table IV.

TABLE III. Sampling.

Number of containers in batch	Number of containers to be sampled
2 – 200	2
Over 200	5

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TABLE IV. Classification of defects.

Test examination	Paragraph	Defects classification
Purity	3.1	Major
Melting point	3.1	Major
Acetone insolubles	3.1	Major
Magnesium content	3.1	Major
Total halide content	3.1	Major
Infrared spectrum	3.2	Major
Workmanship	3.4	Major

4.4.4 Inspection equipment. Commercially available inspection equipment shall be used for all examinations and tests specified in 4.4 and 4.5. The contractor is responsible to have the necessary equipment available, use it correctly and to assure that proper calibration procedures are followed. Government approval of all inspection equipment is required prior to its use for acceptance purposes (see 6.4).

4.5 Test methods and procedures. All tests given in this section shall be performed using prescribed procedures for replicate determination given in standard analytical chemistry textbooks. Also, unless otherwise specified herein, all chemicals and reagents used in the tests shall be ACS Grade or Reagent Grade. Purified and distilled water in accordance with ASTM D 1193 shall be used as required in all test procedures.

4.5.1 Purity. The purity of triphenyl bismuth shall be determined as follows:

4.5.1.1 Apparatus.

a. Liquid chromatograph equipped as follows:
    1. Detector: Ultraviolet (UV) variable wavelength with LC flow-through assembly
    2. Column: Cl8 bonded reversed phase, 5-10 micrometers, 30 cm in length
    3. Pump: Capable of operating up to 4000 psi pressure
b. Laboratory glassware

4.5.1.2 Reagents and standards.

a. Triphenyl bismuth: recrystallized from heptane to maximum absorbance
b. Triphenylmethane: reagent grade
c. Water: High pressure liquid chromatography (HPLC) grade
d. Methanol: HPLC grade
e. Acetone: UV grade
f. Acetonitrile: HPLC grade

4.5.1.3 Procedures.

a. Prepare a stock solution of internal standard triphenylmethane (TPM) by weighing approximately 0.5 g TPM into a 100-mL volumetric flask and diluting to the mark with 50 mL acetone and 50 mL acetonitrile.

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b. Weigh 0.5 g of TPB sample to be analyzed to the nearest 0.1 mg into each of three volumetric flasks.
c. Weigh approximately 0.5 g of TPB standard to the nearest 0.1 mg into each of two 1000-mL volumetric flasks.
d. Accurately pipette into each of the five flasks 1.0 mL of the internal standard stock solution. Dilute the flasks to the mark with acetonitrile.
e. Analyze the three samples and two standard solutions using liquid chromatograph instrument conditions as follows:

1. Detector: Set at lowest wavelength below 226 mm at which a stable baseline can be obtained.
2. Eluent: 80% methanol / 20% water.
3. Flow rate: Approximately 2.0 mL/min
4. Injection: Use the same injection volume for sample and standard solutions.

f. Determine the areas under both TPM and TPB peaks for each sample and standard.
g. The internal standard corrected response factors for the standards should agree within 2 parts per hundred. If not, repeat the test. Calculate the average response factor:

$$\text{Average response factor (R)} = W_1 \text{ x } \left(\frac{I_1}{A_1}\right) + W_2 \text{ x } \left(\frac{I_2}{A_2}\right)$$

Where: A = area of standard TPB peak
W = standard concentration, mg/mL
I = area of standard TPM peak
1 = standard 1
2 = standard 2

h. Calculate the percent TPB in each sample as follows:

$$\text{Percent TPB} = 100 \text{ x } \frac{R \text{ x } B}{C \text{ x } I_s}$$

Where: R = response factor defined above
B = area of sample TPB peak
I_s = area of sample TPM peak
C = sample concentration, mg/mL

4.5.2 Melting point. The melting point shall be determined as follows:

4.5.2.1 Apparatus.
a. Fisher-Johns melting point apparatus
b. Optical cover glass

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4.5.2.2 Procedure.

a. Press a small sample between two optical cover glasses. Place the glasses (with sample) on the heating block of a Fisher-Johns melting point apparatus.
b. Set temperature control at "30."
c. Observe the sample through the small magnifying glass provided with the equipment.
d. When the sample is seen to soften (begin to melt) note the temperature and record to the nearest 0.1˚C.
e. Continue observing until all solid has turned to liquid. Again note the temperature and record to the nearest 0.1˚C.
f. Report the two temperatures described above as the melting range for compliance. Failure of either or both of the recorded values to meet the requirement shall reject the lot.

4.5.3 Acetone insoluble. Acetone insoluble matter shall be determined as follows:

4.5.3.1 Apparatus.

a. 250-mL beaker
b. Gooch crucible, 40-mL capacity
c. Glass fiber filter (Reeve Angel 934 AH or equivalent, 2.4 cm)
d. Oven, gravity convection, regulated at 100±2˚C

4.5.3.2 Procedure.

a. Weigh to the nearest mg, approximately 5 g of sample into a 250-mL beaker and dissolve in 200 mL of acetone.
b. Filter the solution through a tared Gooch crucible containing the glass fiber filter, rinsing out the beaker with acetone and passing the rinsings through crucible.
c. Wash the residue in the crucible with acetone until the washings are colorless.
d. Dry the crucible at 100±2˚C for one hour, cool in a desiccator, and weigh.
e. Calculate the acetone insoluble matter as follows:

$$\text{Acetone insoluble matter, \%} = 100 \text{ x } \frac{B - A}{W}$$

Where: A = tare weight of Gooch crucible, g
B = weight of crucible and residue, g
W = weight of sample, g

4.5.4 Magnesium content. The magnesium content shall be determined as follows:

4.5.4.1 Apparatus.

a. Atomic absorption spectrometer, Varian model 1200 or equivalent
b. Hollow cathode lamp, magnesium
c. Burner, 4-inch single slot for air-acetylene

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4.5.4.2 Reagents.

a. Nitric acid, reagent grade

b. Perchloric acid, reagent grade

c. Hydrochloric acid reagent grade.

d. Magnesium standard solutions – Weight 1.6920 g of magnesium oxide, reagent grade, into a 1-Liter volumetric flask, dissolve in 5 mL of hydrochloric acid and make to volume with distilled water. One mL of this solution contains 1000 micrograms (µg) of magnesium. Transfer 10 mL of the solution into a 100-mL volumetric flask and make to volume. One mL of this solution contains 100 µg of magnesium. Prepare working standards daily containing 0, 3, and 6 µg/mL magnesium by placing appropriate aliquots of the 100 µg/mL magnesium into 100-mL volumetric flasks. Add 50 mL of a 2000 µg/mL strontium solution (prepared by dissolving 4.83 g $Sr(NO_3)_2$ in 1 liter distilled water) to each 100-mL volumetric flask so that the magnesium standards contain 1000 ppm Sr.

4.5.4.3 Standardization.

a. Determine absorbance value for each standard solution using the following atomic absorption conditions:

i. Wavelength: 285 nm

ii. Slit setting: 0.5 nm

iii. Light source: Hollow cathode lamp, magnesium

iv. Flame: Air-acetylene

b. Run a reagent blank.

c. Prepare a calibration curve by plotting the absorbance corrected for the reagent blank reading against the concentration of magnesium for each standard.

4.5.4.4 Procedure.

a. Weigh 0.6-0.8 g of triphenyl bismuth into a 25-mL volumetric flask.

b. Add 5 mL of nitric acid and let the sample digest for 5 minutes at room temperature.

c. Add 5 mL of perchloric acid and carefully digest the sample with heat as needed. The digestion is complete when the solution is a light yellow color.

d. Cool and add 12.5 mL of 2000 µg/mL Sr solution. Dilute to volume with distilled water.

e. Determine the absorbance value using the same conditions as for the standards.

f. Run a reagent blank the same as the samples and subtract this intensity from that of each sample.

g. Determine the concentration of magnesium from the calibration curve.

h. Calculate the ppm magnesium as follows:

$$\text{ppm magnesium} = \frac{(\text{Observed } \frac{\mu g}{mL} \text{ in sample solution}) \times 25}{\text{sample weight, g}}$$

4.5.5 Total halide content. The total halide content shall be determined as follows:

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4.5.5.1 Apparatus.

a. Laboratory glassware
b. Magnetic stirrer

4.5.5.2 Reagents.

a. Nitric acid (reagent grade)
b. Acetone (technical grade)
c. Chloride standard solution – Weigh 0.165 g of sodium chloride into a 1-L volumetric flask, dissolve and make to volume with distilled water. One mL of this solution contains 0.1 mg of Cl.
d. Silver nitrate solution – Dissolve 1.7 g of silver nitrate in 100 mL of distilled water to prepare an approximate 0.1N solution.

4.5.5.3 Procedure.

a. Weigh 0.20 g of triphenyl bismuth into a 50-mL beaker.
b. Mix 25 mL distilled water with 2 mL acetone and add to the sample.
c. Mix on a magnetic stirrer for 15 minutes.
d. Filter the sample solution into a 50-mL beaker.
e. Add 1 mL of nitric acid and 1 mL of silver nitrate solution.
f. Prepare a blank using 1 mL of standard Cl solution and compare with the sample.
g. After 5 minutes, if the turbidity of the sample is less than or the same as the blank, report the sample as containing not more than 0.05% total halides as chloride. If the turbidity of the sample is greater than the blank, report the sample as containing more than 0.05% total halides as chloride.

4.5.6 Infrared spectrum. Verify by examination that an infrared spectrum collected in accordance with ASTM E 168 matches a reference standard of triphenyl bismuth.

4.5.7 Workmanship. Visually examine the sample contamination or foreign materials.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point' packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

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(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The triphenyl bismuth is intended for use in the M864 and XM1128 propellants.

6.2 Acquisition requirements. Acquisition documents should specify the following:
a. Title, number, and date of this specification.
b. Issue of ASSIST to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. Whether a first article sample is required (see 3.2).
d. Packaging requirements (see 5.1).

6.3 Submission of inspection equipment designs for approval. Submit equipment designs, as required, to the Technical Agency (see 6.5)

6.4 Approval of equivalent test methods. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the contracting officer to the Technical Agency (see 6.5). This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.5 Technical agency. Picatinny Arsenal is the technical agency that is responsible for the preparation and revision of the specification. All correspondence to the Technical Agency should be submitted through the contracting officer to: Commander, CCDC – AC, ATTN: FCDD-ACE-QMG, Picatinny Arsenal, NJ 07806-5000.

6.6 Subject term (key word) listing.

Igniter
Base bleed
Propellant
Burn rate modifier

6.7 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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